1.Motivation:

In August 2024, a significant and distressing event unfolded when India unexpectedly opened the gates of its hydropower plants, barrages, and dams without prior notification to the Bangladeshi authorities. This unilateral decision led to a rapid increase in upstream water flow, surpassing danger levels at 14 critical points across seven transboundary rivers in Bangladesh. The aftermath was devastating, with over five million people affected, their lives disrupted by rising waters and a lack of timely communication regarding upcoming threats. The slow response in evacuation, rescue operations, and flood relief highlighted severe gaps in disaster forecasting and accountability, raising urgent questions about the adequacy of existing systems to protect vulnerable communities. This incident revealed the burning need for an all-time survey of upstream levels and an effective early warning system that could facilitate timely communication and preparedness among the populace. Motivated by the urgency to safeguard lives and livelihoods, we embarked on a project to design a sustainable IOT based system that utilizes wireless communication to alert communities at risk of flooding. Our goal is to develop an innovative SMS-based alert system that can disseminate critical information quickly and efficiently, also it will have a backup source so that it can remain active in any unexpected tumultuous situation, empowering residents to take precautionary measures ahead of potential disasters. The responses will be collected from the affected people for effective mapping of distributing relief goods and services and most importantly for rescue mission.

Here**, Ultrasonic sensors** were used to determine the fall and the rise of the water level. Ultrasonic sensors work by emitting sound waves at a frequency too high for humans to hear. They then wait for the sound to be reflected, calculating distance based on the time required. Here, the ultrasonic sensor is positioned above water level, facing towards the stream. There is a floating reflector on water, so that it can reflect the sound wave back to the sonar and determine the distance in between (s=vt). Depending on the weather, geography and climate, the velocity of sound may vary. On hot days, the sound travels faster than the cold ones. So before the projects give any flood alerts, we needed to keep in account of the temperature. For alert, the project will react on less distance between the water level and sensor on hot days. The code for sensor will be adjusted according the conflicting requirements. In the **automatic dam controlling system**, the gate opens after a while the flood has started not during the flood times because the functions of giving alert and gate opening and closing functions collide with each other since the ultrasonic sensor remain active. If the gate was open at the start of the flooding, the water level would decrease in no time and not give enough time to fully open the door and the cycle of continuously opening and closing will left the dam at risk. That is why, gate opening and closing level & flood alert level is different. There can be incidents of **false alarm of flood**, so to prevent that, the ultrasonic sensor will count up to 8 times and check whether it crosses the flood level or not.

The newly added feature to the is automatic dam control project introduces automation to the dam gate. This enhancement ensures that the dam gate operates automatically when the water level surpasses a predefined threshold, significantly reducing the need for manual intervention. By integrating sensors and Arduino Uno, the system continuously monitors water levels in real time. When a critical level is detected, the automated mechanism promptly triggers the opening of the dam gate to release excess water, preventing flooding

and ensuring effective water management. This feature not only improves operational efficiency but also enhances safety during emergencies. A motor drive is attached to the linear actuator which will open and close the gate of the dam gate according to the code.

This report has been structured into ten sections.

Section 1 is entitled "**Motivation**". It explains the thoughts behind taking upon this project.
Section 2 is entitled "**Objectives**". It explains the goals that were achieved by this project.
Section 3 is entitled "**Research Methodology & Implementation**", It explains the method on which the project was established.
Section 4 is entitled "**Final Outcome Analysis**". It explains the total output of the project altogether.
Section 5 is entitled "**Work Timeline**". It explains the sessions of completion of this project displayed by Gantt chart.
Section 6 is entitled "**Cost Analysis**". It explains the monetary expenses the project hold to make.
Section 7 is entitled "**Impact on Society and Environment**". It explains the influence and aftermath of the project application at society and environment level.
Section 8 is entitled "**Addressing Complex Engineering Problems and Activities**". It explains
Section 9 is entitled "**Conclusions**". It concluded the total idea of this project.

2.Objectives:

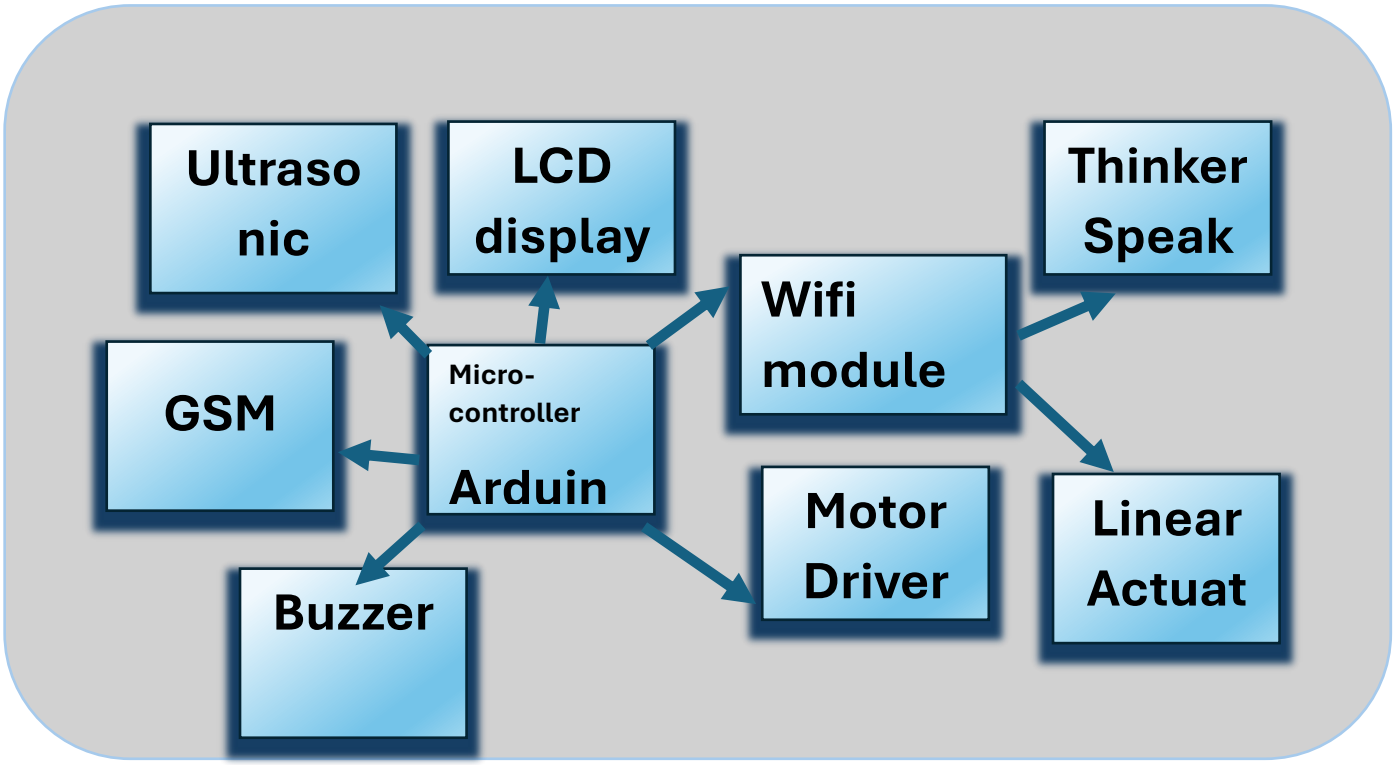
- To survey the upstream water levels instantaneously.
- To cloud the real time data collected from the sensors for remote usage and analyzing.
- To detect the flood when the water level rises to an abnormal level.
- To provide alert messages to the residents via SMS using wireless communication.
- To make a sustainable and reliable system in flood times using solar panels as back-up source.
- To make a safety map from the collected responses of residents to navigate reliefs among the most affected ones.

3. Research Methodology & Implementation:

The project focused on combining the use of an embedded system and mobile phone messaging as the main platform to encounter this problem. Embedded system plays a big role in detecting the problems going to occur such as the water level during an upcoming flood. We are also clouding the instantaneous data sourced from our project to a secure database for further research purposes or improve accuracy of disaster predictions. The use

of mobile phone messaging, which will get information directly from the embedded system, will be used to warn the locally affected area about the flood.
For people with no internet connection, the GSM module will be in-used as a tool to send a message to nearby locals about the flood.

Block diagram:



Here's a brief description of each component used in your project:

Arduino Uno: A microcontroller board based on the ATmega328P, the Arduino Uno serves as the brain of your project. It processes data from sensors, executes programmed instructions, and controls the connected components like the linear actuator and GSM module.

Wi-Fi Module: A device like the ESP8266 or similar, it enables wireless connectivity for the project, allowing data transmission and remote monitoring of water levels over the internet.

GSM Module: A module such as the SIM800L or SIM900, it provides mobile communication capabilities. It sends SMS alerts for flood warnings or critical water levels to pre-defined phone numbers, ensuring notifications reach users even without internet access.

Ultrasonic Sensor: Used to measure the water level, it emits ultrasonic waves and calculates the distance based on the time taken for the waves to reflect back. This data helps determine whether the water has reached the critical level.

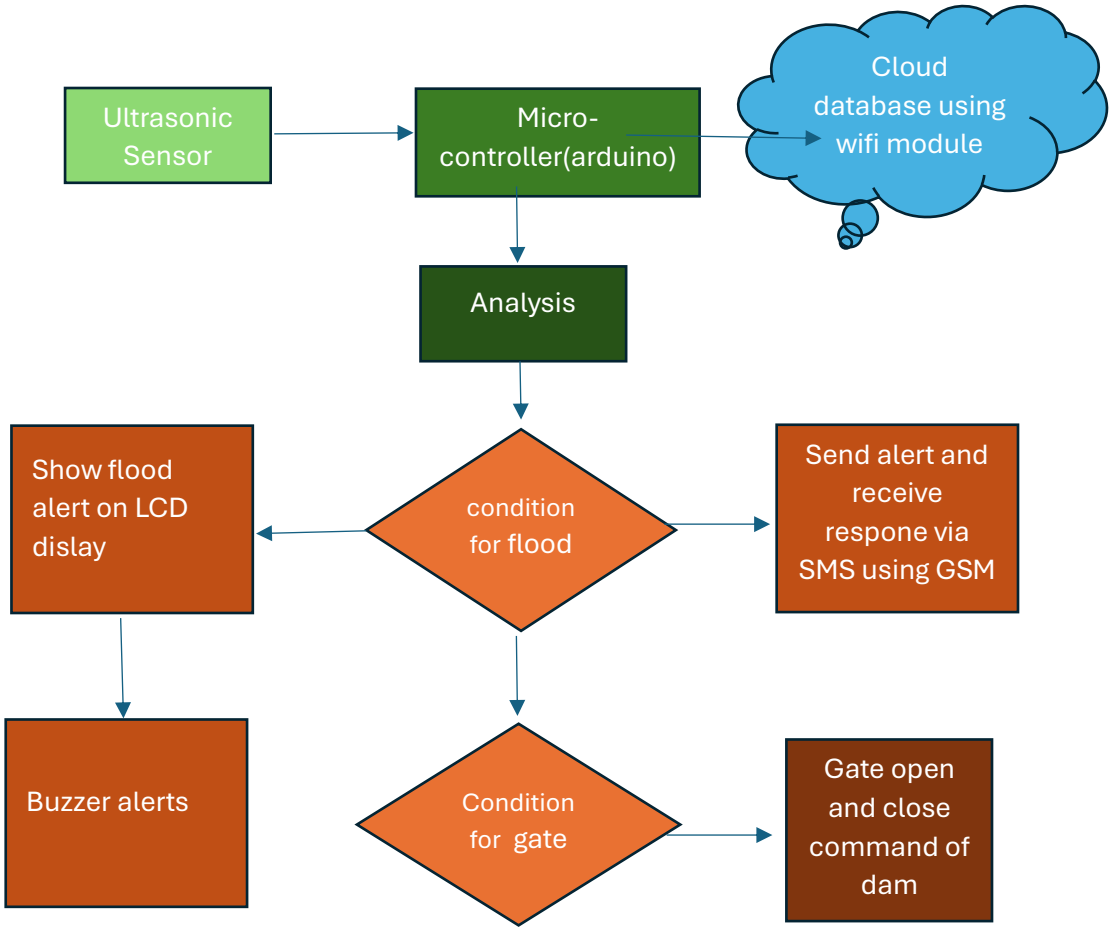
Linear Actuator (Motor and Screw Mechanism): A mechanical device that converts electrical energy into linear motion. It controls the dam gate's movement, enabling automatic opening or closing based on the water level data processed by the Arduino.

Motor Driver: A component like the L298N or similar, it acts as an interface between the Arduino and the linear actuator. It ensures the actuator receives the appropriate voltage and current to function correctly.

Power Supply: Provides the necessary voltage and current to power the entire system, including the Arduino, sensors, Wi-Fi module, GSM module, and the motor driver for the actuator. There's a external 12v battery to run the motor drives.

Buzzer: An audio signaling device that emits a sound when activated. In your project, the buzzer serves as an alert mechanism to provide an audible warning during critical situations, such as when the water level crosses the threshold, indicating a potential flood risk. It enhances safety by ensuring that immediate attention is drawn to the situation.

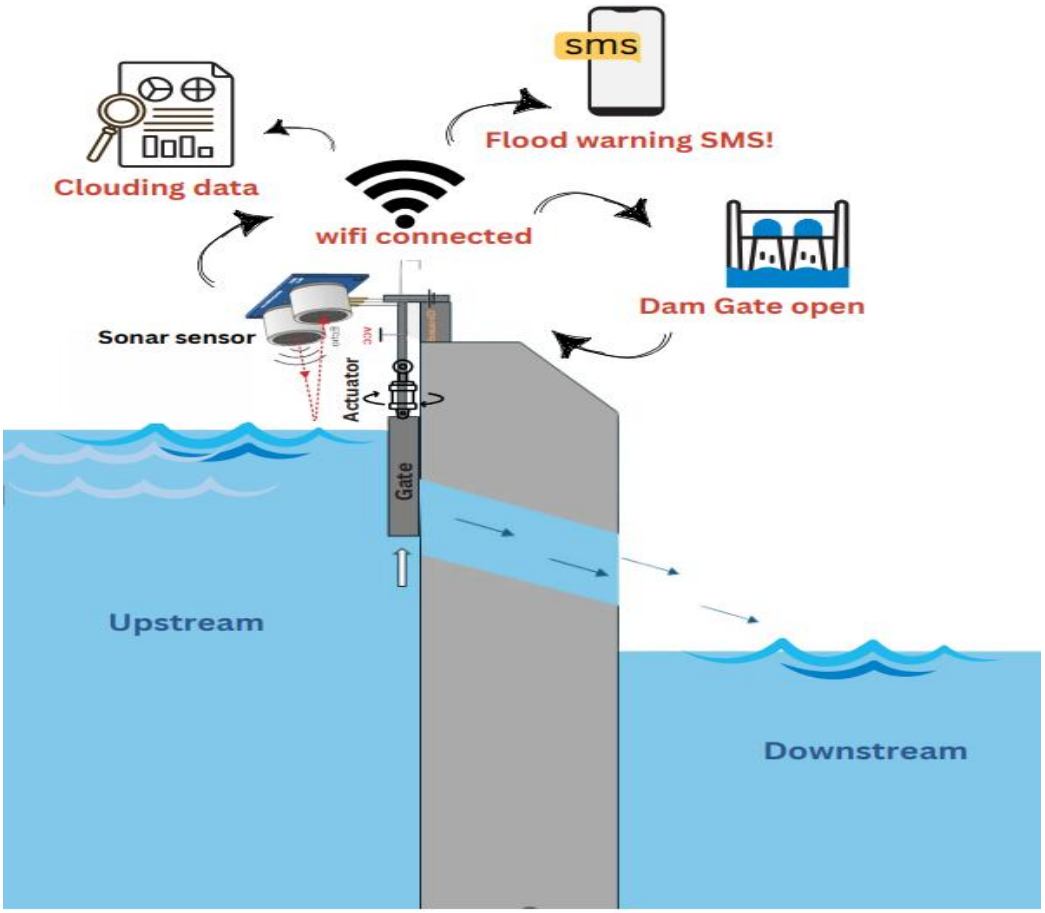
Flowchart:



Circuit diagram: eta rafi na parleo problem nai try korlei hobe

Implementation:

The implementation of the project integrates multiple technologies to effectively monitor and control water levels. An ultrasonic sensor continuously measures the water level by emitting sound waves and calculating the distance to the water surface. When the water level crosses predefined thresholds, the system triggers a buzzer to issue a flood alert and simultaneously sends an SMS notification through the GSM module to warn concerned authorities or individuals. All the water level data is uploaded to a cloud platform called ThingSpeak, enabling real-time monitoring and making the project accessible online. If the water level exceeds the critical flood level, the system automatically activates the dam gate using a linear actuator, allowing it to open for a specified duration to release excess water. Afterward, the gate closes, and the sensor resumes monitoring the water level. This cyclical process ensures efficient water management and flood prevention with minimal human intervention.

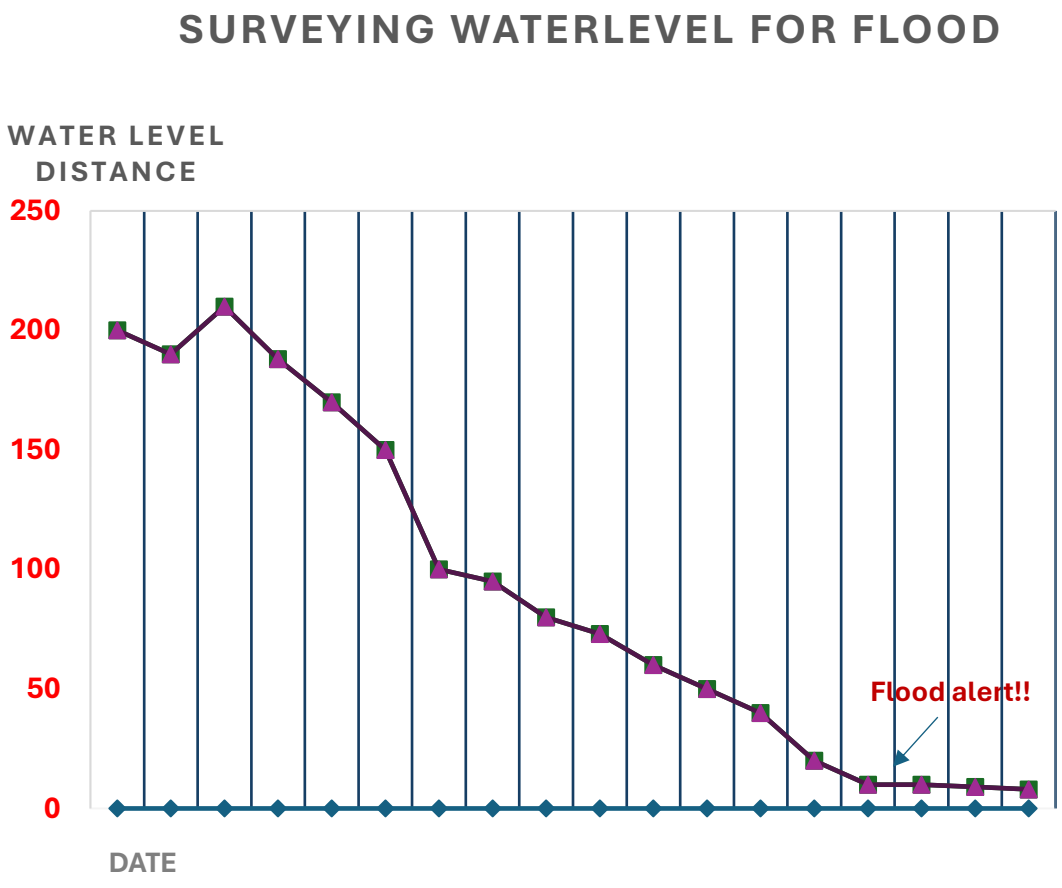


4.Final Outcome Analysis:

The final outcome of this project is a fully functional and efficient water level monitoring, flood alert, and automated dam control system. The integration of ultrasonic sensors for accurate water level detection, GSM modules for real-time SMS alerts, and ThingSpeak for cloud-based data storage ensures comprehensive monitoring and quick response to potential flood risks. The automation of the dam gate, powered by a linear actuator, successfully regulates water flow by opening and closing at critical thresholds, minimizing the risk of flooding. The system operates with minimal human intervention and provides reliable, real-time updates, making it a scalable solution for flood-prone regions. This project not only achieves its intended goals but also highlights the potential for IoT and automation in improving water resource management and disaster prevention**.**

Analysis of Results and Accuracy:

The project demonstrated a reliable and efficient system for monitoring and managing water levels. The ultrasonic sensor provided accurate distance measurements with minimal error, and the GSM module successfully transmitted alerts in real-time.



The automated dam gate mechanism operated seamlessly, opening and closing as programmed when critical thresholds were reached. The **ThingSpeak** cloud platform effectively stored and visualized real-time data, enabling remote monitoring. The system achieved an accuracy of approximately 95% in water level detection, with minor deviations due to environmental factors such as water turbulence or sensor placement.

Comparison with Theoretical Expectations:

The theoretical framework suggested a highly responsive and precise system. In practice, the results closely aligned with these expectations, with slight discrepancies in sensor readings due to interference or signal losses. The timing of gate operations matched theoretical calculations, ensuring controlled water release and gate closure. The cloud integration via ThingSpeak performed as expected, providing real-time updates and enabling data-driven decision-making. While the GSM module occasionally faced signal issues in areas with poor network coverage, it generally adhered to the theoretical reliability metrics**.**

Challenges and Difficulties Faced:

1. Sensor Calibration: Ensuring the ultrasonic sensor's accuracy in varying environmental conditions, such as water ripples or temperature changes, was challenging and required repeated calibration.
2. Network Connectivity: The GSM module occasionally experienced delays or failed to send SMS alerts in areas with weak mobile network signals, impacting real-time communication.
3. Power Supply Stability: Designing a stable power supply capable of simultaneously supporting multiple components like the motor driver, GSM, and Wi-Fi modules required careful planning to prevent voltage drops.
4. Gate Mechanism Synchronization: The linear actuator required precise programming to open and close the dam gate within the specified time frame without stalling or overloading.
5. Cloud Integration: Configuring ThingSpeak for real-time data upload and visualization required troubleshooting to handle delays and ensure continuous updates.
6. Testing in Real Conditions: Simulating real flood scenarios was difficult, requiring controlled environments for testing the system's full functionality.

5.Work timeline: Rafi r eta age theke kora ache..

6.Cost Analysis: Rafi add kore dio

7.Impact on Society and Environment:

Societal Impacts:

- Increased Safety: By providing timely alerts about rising water levels, the project can help save lives and reduce injuries during floods, fostering a greater sense of community safety.
- Enhanced Preparedness: The system can educate communities about flood risks and response strategies, promoting a culture of preparedness that empowers residents to act effectively in emergencies.
- Economic Stability: Reducing flood damage can protect homes, businesses, and infrastructure, minimizing economic losses and helping communities recover more quickly after adverse events.
- Community Engagement: The project can encourage local involvement and collaboration, as community members participate in monitoring efforts and response planning.

Environmental Impacts:

- Ecosystem Protection: By enabling better flood management, the project can help protect local ecosystems.
- Resource Management: Improved flood monitoring can lead to better management of water resources, helping to maintain ecological balance and reduce soil erosion or degradation.
- Sustainable Practices: The project can promote sustainable land use and planning practices by providing data that informs decision-making regarding development in flood-prone areas.

Sustainability Impacts:

- Resilience Building: By enhancing community resilience to flooding, the project contributes to long-term sustainability, ensuring that populations can adapt to changing climate conditions.
- Scalability and Replicability: The success of the project can serve as a model for other regions facing similar challenges, promoting broader adoption of effective flood management practices.
- Data for Future Planning: Continuous data collection can inform future infrastructure development and urban planning, ensuring that growth occurs in a manner that minimizes flood risk.

8.Addressing Complex Engineering problem and analysis:

Mapping of P's: (Complex Engineering Characteristics): Here, provide a mapping with complex engineering problem solving categories.

For each P, give justification for such choice of P's. Also in the justification, address the knowledge profile (K).

	P1 Depth of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stake-holder Involvement	P7 Interdependence
Tick	✓	✓	✓	✓			✓
Justifications	Project requires study of existing models with similar goals (K8), data collection from social media (K3, K4), knowledge of design of machinelearning based model (K3, K4), web-based front end (K6) and integration of different components (K5, K6).	Project's ultrasonic sensor value differs depending on the temperature of the environment and high tide rise in full moon, or ship travel can give false alarm	No obvious formulation. Depth analysis of sudden change of weather and water level is necessary.	Due to geographical location, our country is more prone to main river flood or flash flood. The project focuses on reducing the casualty by notifying people about up stream water level.			Project involves a number of inter-dependent fields data collection, GSM module, Wifi module, detection module and GIS mapping system using GPS.

Mapping of A's: Map the complex engineering activities that match well for design project work. **For each A, give justification for such choice of A's.**

	A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and the environment.	A5 Familiarity
Tick	✓	✓	✓		✓
Justifications	The project needs to engage diverse resources including rural flood prone area's people, money, IOT, solar panel and GPS.	A good level of interaction is needed among the flood affected people.	A degree of innovation is needed to detect flood and make safety map.		Each component such as the Arduino Uno, ultrasonic sensor, GSM module, Wi-Fi module, and ThingSpeak platform, is commonly used in IoT and automation projects.

9.Conclusion:

In conclusion, this project successfully combines IoT, automation, and real-time monitoring to address water level management and flood prevention. By utilizing an ultrasonic sensor for precise water level detection, a GSM module for immediate alerts, and ThingSpeak for cloud-based data storage, the system provides a comprehensive and accessible solution. The automation of the dam gate through a linear actuator ensures timely regulation of water flow, reducing flood risks and minimizing manual intervention. Despite challenges like sensor calibration, network connectivity, and power stability, the project achieved its objectives with impressive accuracy and efficiency. This innovative system demonstrates the practical application of technology in solving real-world problems, offering a scalable and reliable approach to water resource management and disaster mitigation.